INSTITUTO POLITÉCNICO NACIONAL
ESCUELA SUPERIOR DE CÓMPUTO

Selected Topics in Cryptography

Lab Session 1: Non-singular elliptic curves *February 19, 2025*

Please solve the following exercises by your own.

1. Algorithms

1. Write the pseudocode of an algorithm that receives as input, the parameters of a non-singular elliptic curve a, b, p, where $p \geq 3$ is a prime number. The output must be the points that belong to curve, using a notation of three coordinates, i.e the point of infinity $\mathcal{O} = (0, 1, 0)$ and for any other point $P \neq \mathcal{O}$, $P = (x, y, 1)$. Please use the notation for pseudocode given in class.

2. Programming exercises

You can use C/C++, Java or Python to develop your source code

1. Design a function that receives as input a prime number $p \geq 3$ to find the quadratic residues modulo p and also the square roots modulo p. For example if $p = 11$ and we know that $2^2 \bmod 11 = 4$ and $9^2 \bmod 11 = 4$ we know that 4 is a quadratic residue and its square roots are 2 and 9.
2. Design a function that receives as input a prime number $p \geq 3$, and generates a non-singular elliptic curve over $\mathbb{Z}_p$, ie. your function must output the parameters a, b such that the discriminant is different from zero.
3. Design a function that receives as input the parameters of a non-singular elliptic curve E, given by a, b, a prime number $p \geq 3$ and a point of three coordinates $P = (x, y, 1)$ where $x, y \in \mathbb{Z}_p$ or $\mathcal{O} = (0, 1, 0)$. The output of your function will be true if $P \in E$ or false if $P \notin E$

3. Products

- You must write a brief report, containing:
  1. Your personal information, date of the lab session and the topic that we are studying in this lab session.

2. Pseudocode for the algorithm that you designed in section 1.
3. The source code for points 2.1 , 2.2, 2.3. Please include a brief paragraph explaining your functions.
4. At least 5 examples that you use to test each function of section 2.
5. Screen shots of your programs running.

4. Evaluation

- Advances in class: 2 points
- Source code: 3 points
- Program running: 3 points
- Report: 2 points